



AI reinvention made real

Accenture Innovation Challenge 2026

Team details

TEAM NAME: Synergy



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Problem	Solution	Impact
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Problem	Emergency triage relies on a point-in-time assessment while patient conditions & ED capacity change. This can cause missed deterioration, mis-prioritization & avoidable waiting
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WHY THE ED IS HARD TO MANAGE



HIGH DEMAND

Large and unpredictable patient inflow



LIMITED CAPACITY

Beds, staff and diagnostic resources are finite



TIME-CRITICAL

A delay matters disproportionately for high-acuity patients

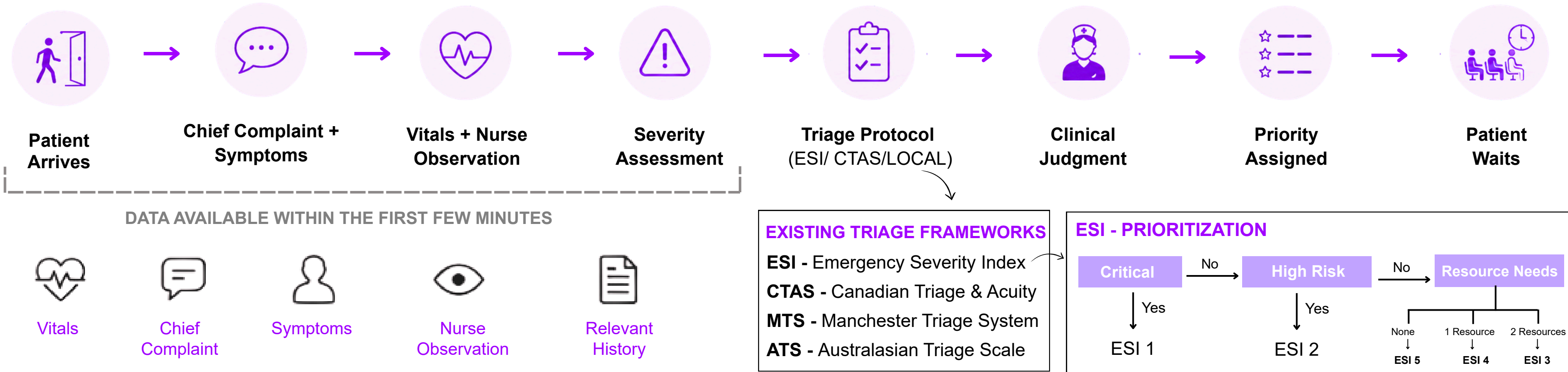


PRIMARY USER - TRIAGE NURSE

- Assesses patients at arrival
- Synthesizes symptoms, vitals & observations
- Assigns initial priority
- Reassesses when condition changes

Challenge: High volume • Time pressure • Incomplete information

CURRENT USER FLOW



PAIN POINTS

1. COGNITIVE LOAD



One nurse synthesizes multiple signals under time pressure.

Higher chance that subtle signals are missed.

2. STATIC PRIORITY



The decision is made at T_0 - but the patient exists at T_1 , T_2 , $T_3...$

Deterioration can occur between reassessments

3. CAPACITY BLIND SPOT

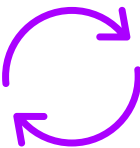
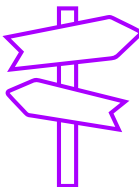


Clinical priority does not automatically translate into the best available care stream.

4. REASSESSMENT GAP



A deteriorating patient may need a new assessment before the original priority is revisited.

	Problem		Solution			Impact		
Medicine Informed Artificial Intelligence for smarter Triage : An end-to-end , human in the loop platform that augments clinical judgement to optimally prioritize and route patients								
1. Data Intake	 Structured Vitals HR, BP, RR, SpO ₂ , Temp, Pain score	 Chief Complaint Free-text or symptom list	 Demographics Age, Sex, Arrival mode	 History Allergies, conditions, meds, prior visits (EHR) if available	 Observation Mental status, distress, mobility, appearance	Technical Highlights • Medicine-Informed: Not just data-driven: leverage clinical knowledge, ontologies, protocols and safety rules. • Temporal & Dynamic: Models patient risk trajectory over time, not just a single snapshot. • Uncertainty-Aware: Quantifies what the model knows vs. doesn't know and acts accordingly. • Human-in-the-Loop: AI recommends, clinicians decide. Overrides are captured for learning and audit. • Interoperable: FHIR/SMART on FHIR based integration with EHR, monitors, and hospital systems. • Safe & Governed: HIPAA-aligned, encrypted, auditable, fair, and designed for responsible AI.		
	Data Quality & missing value Handling imputation uncertainty Encodeing							
2. AI processing Core	Symptom & Language Understanding (NLP) • Clinical NLP to extract entities & normalize to SNOMED CT • Negation & uncertainty detection (ConText) • Multilingual translation layer • Red-flag concept recognition via domain lexicons	Risk Scoring Engine • Medicine-informed ML model (XGBoost + Deep Neural Networks) • Trained on de-identified ED datasets aligned to ESI • Captures complex non-linear interactions & atypical presentations • Outputs: Acuity Score (0–5), Confidence (0–100%)	Deterioration Prediction Module • Temporal model (GRU / Temporal CNN) on vitals trends & static risk factors • Predicts probability of clinical deterioration in waiting period (e.g., 0–6 hrs) • Outputs: Deterioration Risk & Time-to-deterioration distribution	Bias & Consistency Check • Disparity analysis across age, gender, ethnicity • Compare AI score vs. nurse initial triage (\$\Delta\$-threshold) • Uncertainty quantification using Monte Carlo Dropout • Flags for second look when discordant or high-risk				
	Feature Store Model registry Versioning Monitoring for Drift & performance							
3. Decision Human interface	 Recommend , Don't Decide Suggested acuity level + rationale, not auto-assignment	 Escalation Triggers Low confidence or red flags → immediate human re-check	 Explainability layer SHAP/LIME explanations highlight key clinical drivers of the score	 Nurse Control : Accept, modify or override with notes (decision always nurse-led)	 Override Logging All overrides & actions captured for audit & continuous learning		KEY OUTPUTS  Dynamic Risk Score Deterioation risk & trend  Acuity (ESI) Recommendation Reassessment Priority	
4. Feedback & Learning Loop	 Outcome Capture Disposition, ICU admission, deterioration, time-to-treatment, LOS, mortality	 Model Retraining & Calibration Periodic retraining, Bayesian calibration, hospital-specific tuning	 Performance Monitoring AUROC, AUPRC, calibration, fairness, alert burden	 Data Pipeline ETL, data quality checks, label engineering, de-identification	 Continuous Improvement: Human-in-the-loop feedback better models, safer care			
	Closed loop learning governance explainablity & audit trails privacy & security by design							

Problem		Solution	Impact				
MODEL IN ACTION SENSE Continuously ingests multi-source data UNDERSTAND Understands risk trajectory & ED context PRIORITIZE Dynamically updates priority as patient & ED state change ROUTE Recommends / executes best available care stream in real-time NURSE IN CONTROL Nurse can accept, modify or override at any point. REASSESS Detects change & triggers review PatientTriage.ai		EVALUATION CRITERIA FOR MODEL Clinical Effectiveness Correctly identify high-risk patients and prevent deterioration. Timeliness Prioritize and route patients quickly and appropriately. Safety & Reliability Minimize adverse events and decision errors across all risk levels. Operational Efficiency Improve throughput and resource utilization in the ED. Fairness & Trust Ensure unbiased, consistent and explainable decisions clinicians trust.					
		KEY METRICS • North star Time to Appropriate Care (TTAC) Median time for high-risk patients to first appropriate care ↓ 22% improvement TRIAGE ACCURACY (ESI 1/2) % of high-risk patients correctly prioritized by AI (recommended or executed) ↑ 18% improvement LEFT WITHOUT BEING SEEN (LWBS) % reduction in patients leaving before evaluation ↓ 28% reduction MEDIAN ED LENGTH OF STAY Overall reduction in patient length of stay ↓ 15% reduction					
AI DECISION LOGIC ESI 1 / ESI 2 (Critical / High Risk) AI RECOMMENDS Final decision by Triage Nurse ESI 3 / ESI 4 / ESI 5 (Urgent / Less Urgent / Non - Urgent) AI EXECUTES Priority & routing executed by AI		BUSINESS IMPACT Better Patient Outcomes Operational Efficiency Cost Savings Scalable & Sustainable Staff Empowerment					

Thank you

